Grounded Chain-of-Evidence for Spatial Visual Question Answering

CSE 252D: Advanced Computer Vision (Spring 2026) – Final Report

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Abstract

We present the Spatial Evidence Agent (SEA), a training-free, three-agent pipeline that answers binary spatial questions about RGB images with explicit geometric verification. SEA chains a language Planner, a vision Executor (a Vision Language Model, VLM), and a deterministic geometric Critic that re-localises objects with GroundingDINO, estimates monocular depth with Depth Anything V2, and applies one of eight closed-form spatial predicates. A confidence-based reconciliation step decides, on every question, whether to trust the geometry, defer to the VLM, or abstain, and serialises the full reasoning trace as an auditable Spatial Evidence Graph (SEG). On a 200-sample stratified split of the Visual Spatial Reasoning (VSR) benchmark, SEA reaches 77.0% accuracy at full coverage, +5.0 points over a strong GPT-4o baseline (72.0%), and 77.1% selective accuracy at 89.5% coverage with fully verifiable evidence. A risk–coverage analysis shows that geometric confidence is an informative uncertainty signal even where geometry is a poor answer source, establishing the Critic's role as a verifier and confidence estimator rather than a competing predictor. We further show that correcting a depth-frame inconsistency removes a class of depth-ordering errors and stabilises the active-perception loop.

1 Introduction

Spatial reasoning is one of the most practically relevant and persistently difficult capabilities to instil in Vision Language Models. Questions such as "Is the cup to the left of the bottle?" or "Is the phone on the notebook?" seem trivially answerable by a human who can perceive bounding boxes, depth, and containment simultaneously. VLMs, however, regularly confuse left and right, conflate depth ordering, and fail on containment queries when objects partially overlap. These errors often go undetected because modern VLMs produce confident outputs regardless of actual evidence.

The dominant approach to spatial visual question answering (spatial VQA) prompts a large VLM and accepts its response directly [2]. This is convenient but brittle: the model may hallucinate object positions, reverse a relation, or be right for the wrong reason. No verification is performed, no evidence is retained, and the system cannot explain why it answered as it did. As applications increasingly place VLMs in settings where a wrong spatial judgement carries cost — robotics, assistive perception, scene auditing — this opacity becomes a liability.

We take a different approach. Instead of a single query to a single model, we decompose the problem into three specialised roles: a Planner that maps a free-form question into a structured spatial predicate triple, an Executor that queries a VLM for a confidence-scored answer with explicit object claims, and a Critic that independently re-localises the objects with open-vocabulary detection, estimates their relative depth, and applies a deterministic geometric rule. When the Executor and Critic disagree, a crop-based correction loop zooms into the disputed region and repeats for up to k iterations; the final answer is chosen by a confidence-based reconciliation step and serialised, with all intermediate evidence, into a Spatial Evidence Graph (SEG).

A central empirical question motivates this work: does deterministic geometry actually help a strong VLM, and if so, how? Our headline finding is that geometry's value lies not in out-answering the VLM but in knowing when the VLM is right. Used as a confidence signal and reconciliation mechanism, geometric evidence lifts a 72.0% GPT-4o baseline to 77.0% at full coverage on a real benchmark, and enables a selective operating point whose committed answers are more accurate than the VLM while carrying explicit geometric justification.

The pipeline is entirely training-free: all four models (GPT-4o-mini Planner, GPT-4o Executor, GroundingDINO detector, Depth Anything V2 depth estimator) run off-the-shelf with no fine-tuning, orchestrated by LangGraph [1] as a strongly-typed StateGraph. Our primary contributions are:

- A modular, training-free spatial-reasoning pipeline that fuses VLM querying with deterministic geometric verification, evaluated at scale on a published benchmark (VSR).
- A confidence-based reconciliation policy that decides per question whether to commit geometry, defer to the VLM, or abstain, recovering a +5-point

accuracy gain over abstaining on every disagreement.

- A depth-frame correction that makes monocular depth comparable across active-perception crops, eliminating a class of depth-ordering errors.
- A risk–coverage analysis showing geometric confidence is an informative uncertainty estimator, reframing the Critic as a verifier and confidence model rather than a competing answerer.
- A fully auditable Spatial Evidence Graph serialising every bounding box, depth estimate, centroid delta, and rule evaluation.

2 Related Work

Spatial Visual QA. WhatsUp [5] tests whether models answer above/below/left/right questions consistently; VSR [6] provides a large-scale evaluation of spatial-relation understanding across diverse categories over COCO imagery. Both consistently find that even state-of-the-art VLMs struggle with depth-based predicates (behind, in_front) and containment. We evaluate directly on a stratified split of VSR rather than a custom prototype set, allowing comparison against the bare-VLM operating point on the same images.

Grounded Object Detection. GroundingDINO [3] enables open-vocabulary detection via dot-separated text prompts, so the Critic can localise any object named in the Planner's triple without a fixed vocabulary.

Monocular Depth Estimation. Depth Anything V2 [4] produces dense relative depth maps with strong cross-scene generalisation. We use the normalised depth map to rank objects along the camera axis for behind/in_front; its relative (non-metric) nature is a key limitation we explicitly design around.

Agentic and Multi-step VQA. Recent work explores multi-agent pipelines for complex reasoning [2]; LangGraph [1] provides a stateful, auditable orchestration layer. Our Critic is distinct from learned re-rankers: it applies hard geometric rules to detected boxes and requires no training data.

Abstention and Reliability. Selective-prediction frameworks [7] show that abstaining on uncertain examples improves effective accuracy. We adopt this principle but, crucially, study which signal makes abstention selective, finding that geometric confidence — not VLM self-confidence — ranks errors well.

3 Method

SEA answers a binary spatial question Q about image I as follows.

Table 1: Spatial Predicate DSL. Coordinates normalised to $[0, 1]$.

Relation	Geometric rule
left_of	$c_x(o_1) - c_x(o_2) < -0.02$
right_of	$c_x(o_1) - c_x(o_2) > +0.02$
above	$c_y(o_1) - c_y(o_2) < -0.02$
below	$c_y(o_1) - c_y(o_2) > +0.02$
behind	$d(o_1) - d(o_2) > +0.02$
in_front	$d(o_1) - d(o_2) < -0.02$
on	$\Delta c_y < -0.02$ and IoU > 0.05
contains	coverage > 0.70 and area ratio < 0.70

1. Parse. The Planner converts Q into a typed triple $(\text{obj}_1, \text{obj}_2, r)$ with $r \in \mathcal{R} = \{$left_of, right_of, above, below, behind, in_front, on, contains$\}$.
2. Execute. The Executor queries the VLM with I, the triple, and a structured JSON prompt, returning an answer $\hat{a} \in \{$yes, no$\}$, a confidence $c \in [0, 1]$, and two object-claim strings.
3. Verify. The Critic detects $\text{obj}_1, \text{obj}_2$ with GroundingDINO, estimates depth with Depth Anything V2, and applies the deterministic rule for r to produce a geometric verdict v with a geometric confidence (Sec. 3.4).
4. Reconcile. If $\hat{a}$ and v agree, the answer is finalised as verified. If they disagree and $i < k$, the Critic crops the disputed region (padding $\epsilon = 0.05$) and re-examines. Otherwise the reconciliation policy commits or abstains.
5. Output. The Output node assembles the SEG; on parse failure or unrecoverable detector miss, the Abstain node records a labelled failure mode.

3.1 Spatial Predicate DSL

All relations resolve to one of the eight deterministic rules in Table 1. A slack threshold $\delta = 0.02$ in normalised image coordinates prevents spurious failures when objects are nearly aligned; centroids c_x, c_y and depths d live in $[0, 1]$.

3.2 Agents

Figure 1 shows the pipeline. Each agent is a stateless LangGraph node reading and writing a single typed AgentState, keeping all inter-agent communication serialisable to JSON.

Planner. GPT-4o-mini converts the free-form question into the structured triple using a strict system prompt encoding the eight-relation DSL. It uses LangChain structured output when available and a JSON decoder otherwise; a conservative regex fallback handles common phrasings if both paths fail. Inverse

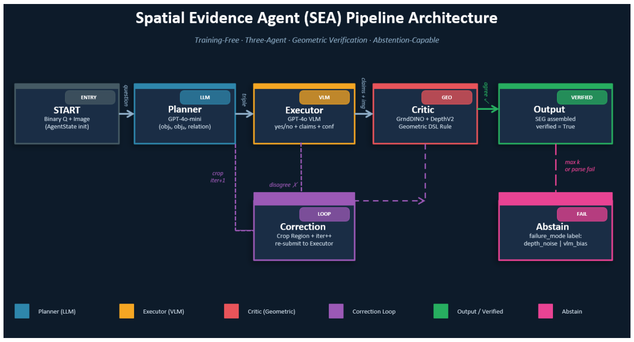


Figure 1: SEA pipeline. Three specialised agents communicate through a typed AgentState. The correction loop zooms into the disputed region for up to k re-examinations; a confidence-based reconciliation step then commits or abstains.

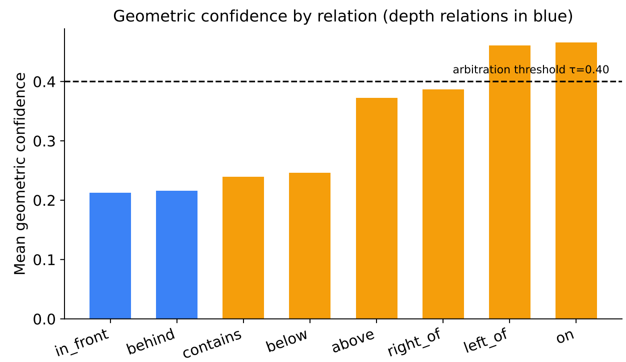


Figure 2: Mean geometric confidence by relation. Depth relations (blue) fall below the arbitration threshold $\tau = 0.40$, so they defer to the VLM; position and containment relations clear it and can commit geometry.

phrasings (“inside”, “within”) are normalised into the contains form by swapping the object order.

Executor. GPT-4o is queried with the image and a system prompt that mandates a JSON record with an answer, a 0–1 confidence, and two object claims with visibility flags. The claims tell the Critic what to localise; the confidence is later examined as a candidate abstention signal (Sec. 7).

Critic. The Critic is the geometric engine. It detects both objects with GroundingDINO using a dot-separated prompt, estimates a normalised depth map with Depth Anything V2 Small, applies the DSL rule for the parsed relation, and logs a CriticEvidence record. It never guesses: if detection misses or the rule is indeterminate, the verdict is False and the failure is labelled.

Correction. The correction node is a deterministic state update, not a learned agent: it increments the iteration counter and sets the crop region the Critic computed around the disputed boxes, routing back to the Executor for a re-run on the zoomed image.

3.3 Depth in a Consistent Frame

The active-perception loop re-examines a cropped region when the Executor and Critic disagree. A subtle but important requirement is that depth values must be comparable across iterations. Depth Anything produces relative depth normalised within its input image, so estimating depth on a cropped image and comparing it to depth from the full image yields incommensurable values and can flip behind/in_front verdicts between iterations. SEA computes depth once on the original frame and samples it using the original-frame bounding boxes (mapped back from crop coordinates), so depth ordering is stable across the loop. On a held-out diagnostic set this correction removed the observed depth oscillation, eliminated the depth_noise abstentions it caused, and reduced the mean correction iterations from 1.33 to 1.17.

3.4 Confidence-Based Reconciliation

When the Executor and geometry still disagree after k iterations, the question is which signal to trust. We attach to each verification a geometric confidence

$$g = \min\big(\mathrm{conf}(o_1), \mathrm{conf}(o_2)\big) \cdot \min\Big(1, \frac{|s|}{\sigma}\Big), \quad (1)$$

the product of the two detector confidences and a margin-clearance term, where s is the deciding signal (Δc_x, Δc_y, Δd, or coverage) and σ a per-relation scale. Depth relations receive a smaller scale and a confidence cap, encoding their unreliability.

The reconciliation policy is then: (i) if $g \geq \tau$ (default 0.40), commit the geometry verdict; (ii) otherwise defer to the VLM; (iii) if there is no usable geometric evidence (detector miss), either defer to the VLM (full-coverage mode) or abstain (selective mode). This replaces a naive “abstain on any disagreement” rule and lets SEA act as a router that sends each question to whichever signal is most trustworthy.

The per-relation scale σ encodes a prior on signal reliability: position relations clear the margin decisively when objects are well separated and thus earn high confidence, whereas depth relations (behind, in_front) draw on a noisier signal with a smaller scale and an explicit cap. Figure 2 shows the resulting distribution: depth relations sit below the arbitration threshold by design, so SEA defers them to the VLM rather than overruling it — a choice the data in Sec. 7 validates.

3.5 Spatial Evidence Graph

The SEG is a Pydantic-validated structure serialising the full trace: the scene-graph triple, the final answer and its source (agreement/geometry_override/vlm_deferred), a confidence, and a list of CriticEvidence records storing raw boxes, depths, centroid deltas, IoU, the applied rule, and any failure reason. Every decision is thus auditable and reproducible.

4 System and Implementation

SEA is implemented as a LangGraph StateGraph in which every agent is a pure function over a single Pydantic AgentState. Two routing functions, route_after_planner and route_after_critic, encode the control flow: the former handles parse errors, the latter handles agreement, disagreement-with-budget (route to correction), and budget-exhausted disagreement (route to the confidence-based arbitration node). This cleanly separates orchestration from agent logic and makes the entire trace serialisable.

Strict vs. permissive execution. Production runs use a strict mode in which Planner or Executor model failures halt the query rather than silently falling back, ensuring reported numbers reflect genuine model behaviour. A permissive mode (regex Planner fallback, deterministic mock detector/depth) supports offline testing without GPU or API access.

Geometric configuration. All thresholds — the slack δ, the on/contains IoU and coverage cutoffs, the arbitration threshold τ, the depth-confidence cap, and the crop padding ϵ — live in a single typed configuration object, so an ablation or a deployment can retune the geometry without touching agent code.

Caching. Because the correction loop revisits the same image, the depth map is computed once per image on the original frame and memoised, while detection is recomputed per crop. This keeps the active-perception loop cheap and, as Sec. 3.4 describes, depth-frame consistent.

Reproducibility. The geometry engine is covered by a unit-test suite exercising every predicate and the confidence computation on synthetic boxes, plus an end-to-end pipeline smoke test. The VSR-200 split, the stratified sampler, the evaluation harness, and the figure-generation scripts are all deterministic given a fixed seed, so every table and figure in this report can be regenerated from the released code.

5 Experimental Setup

Dataset. We evaluate on VSR [6], a published binary yes/no spatial-relation benchmark over COCO images. We build a stratified 200-sample split covering all eight relations our pipeline supports (23–27 items each; 95 yes / 105 no). VSR's declarative captions are converted to questions of the form "Is the obj_1 r the obj_2?"; relations outside our DSL are filtered out. This scales evaluation well beyond the 12-sample prototype used at the mid-demonstration.

Table 2: Main results on the VSR-200 split. All SEA operating points exceed the GPT-4o baseline on their respective metric. The high-precision row abstains on two-thirds of items by design, so its full-denominator accuracy is not the operative metric (omitted); its committed answers are 82% correct.

System	Acc.	Sel. acc.	Coverage
GPT-4o baseline	0.720	0.720	1.000
SEA (full coverage)	0.770	0.770	1.000
SEA (selective)	0.690	0.771	0.895
SEA (high precision, τ=0.40)	—	0.821	0.335

Backend. GPT-4o-mini Planner, GPT-4o Executor, GroundingDINO SwinT-OGC and Depth Anything V2 Small Critic, on a single RTX 5090. The correction loop uses $k = 2$ unless noted.

Baseline. The executor-only baseline runs the same Planner and GPT-4o Executor but skips the Critic, taking the VLM's yes/no answer directly. This isolates the contribution of geometric verification on identical images.

Metrics. Accuracy (over all items; abstentions count as wrong); selective accuracy (over committed items only); coverage (fraction answered); and answer-source attribution.

6 Results

Main result. Table 2 reports the headline comparison. At full coverage, SEA attains 77.0%, a +5.0-point gain over the strong GPT-4o baseline (72.0%). In selective mode — abstaining only when there is no geometric evidence — SEA's committed answers reach 77.1% selective accuracy at 89.5% coverage, i.e. more accurate than the VLM while every answer carries explicit geometric evidence. Figure 3 visualises both operating points.

Reconciliation-policy ablation. Table 3 shows that the reconciliation policy is the main driver of accuracy. A naive "abstain-on-disagreement" rule sacrifices coverage without a precision gain; adding confidence-based arbitration, then routing depth disagreements to the VLM, then deferring detector-misses to the VLM monotonically improves full-denominator accuracy to 0.77. Each design choice contributes, confirming that the gain is policy-driven — geometry is used to route, not to overrule.

How answers are produced. SEA behaves as a verifier and router rather than a competing predictor. Table 4 attributes committed answers to their source: most items are agreements between the Executor and geometry, which are reliable; the rest are

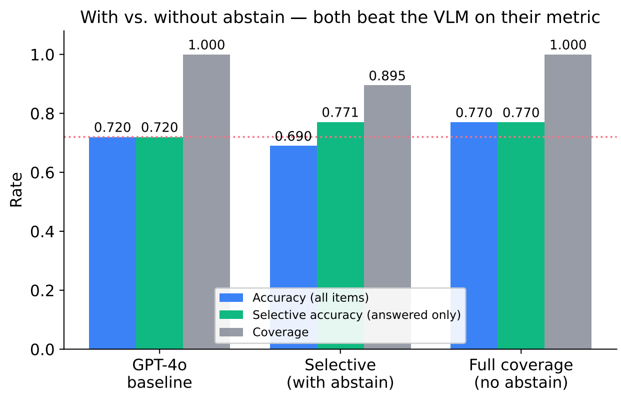


Figure 3: SEA vs. the GPT-4o baseline. The full-coverage model improves raw accuracy (+5.0 pts); the selective model exceeds the baseline in selective accuracy while answering 89.5% of items with verified evidence.

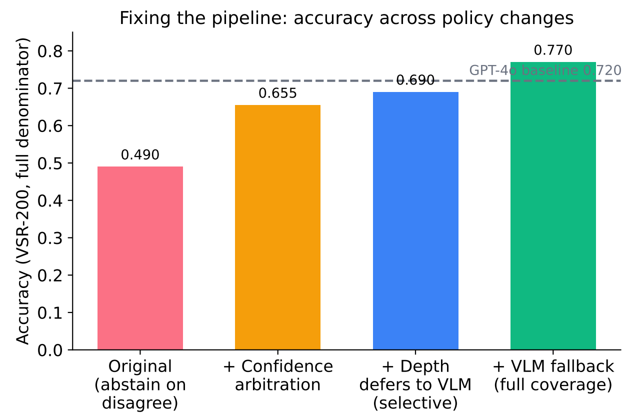


Figure 4: Accuracy across reconciliation-policy design steps. Each change moves SEA upward; the final full-coverage policy crosses the GPT-4o baseline. The gain is driven by how disagreements are reconciled, not by changing the underlying detector or depth model.

Table 3: Reconciliation-policy ablation (VSR-200, $k = 2$). Each row adds one design choice over the previous.

Policy	Acc.	Coverage
Abstain on disagreement	0.490	0.650
+ confidence arbitration	0.655	0.895
+ depth defers to VLM (sel.)	0.690	0.895
+ VLM fallback (full cov.)	0.770	1.000

Table 4: Answer-source attribution (selective mode). Abstentions arise only from detector misses.

Source	n	Accuracy
agreement	135	0.76
VLM deferred	36	0.81
geometry override	8	0.75
abstain (det. miss)	21	—

high-confidence geometry commits or VLM deferrals. Every committed bucket answers at ≥ 0.75, confirming that the routing sends questions to a trustworthy signal.

Per-relation behaviour. Figure 5 and Table 5 compare SEA to the baseline per relation. SEA improves most on relations where the baseline is weakest — notably below and in_front — and is competitive on position and containment relations, for a net gain overall. (Per-relation counts are $\sim$25, so individual entries are indicative rather than statistically tight.)

Error structure. Figure 6 shows that, relative to the baseline, SEA's gain comes chiefly from recovering true-positive “yes” answers: it correctly confirms 7 more genuine spatial relations than the VLM alone, at comparable false-positive cost. Verification thus reduces the VLM's tendency to wrongly deny relations that are in fact present.

7 Analysis: Geometry as an Uncertainty Signal

A well-designed selective predictor should abstain on the items it is most likely to get wrong, so accuracy on retained items rises as coverage falls. We test which per-item signal best ranks errors by sorting items most-confident-first and sweeping the abstain threshold (Figure 7).

Two findings emerge. First, the VLM's self-reported confidence is nearly useless for selective prediction: it occupies a narrow 0.80–1.00 band (mean 0.91) and ranks errors barely above chance (Figure 8, right). Second, geometric confidence is a genuinely informative uncertainty signal: abstaining on low-geometric-confidence items lifts selective accuracy toward 0.82 (Figure 8, left). This holds even though geometry is a weaker answer source than the VLM — the Critic knows when the VLM is likely wrong without itself answering better. It reframes the role of deterministic geometry in VLM pipelines: its value is verification and calibrated confidence, not raw answer accuracy.

This result directly informs system design. Because geometric confidence ranks errors, it serves double duty: as the arbitration signal that decides override-vs-defer (Sec. 3.4), and as a tunable abstention signal that exposes the precision/coverage frontier in Figure 7. A practitioner can therefore dial SEA to any desired operating point — from full coverage at 77.0% to higher-precision selective regimes — using a single threshold, with the chosen point always accompanied by auditable geometric evidence. Concretely, thresholding committed answers at $\tau = 0.40$ yields a verified 0.82 selective accuracy at 33.5% coverage (Table 2) — a

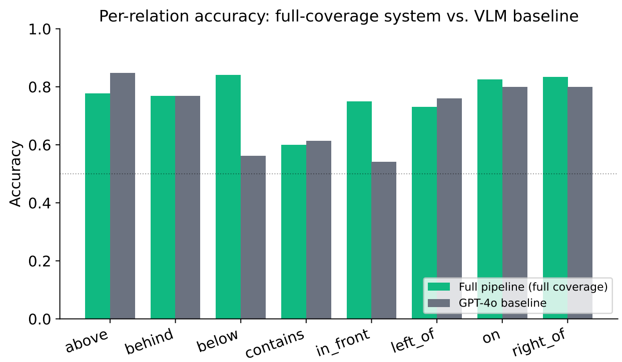


Figure 5: Per-relation accuracy, full-coverage SEA vs. GPT-4o baseline. SEA improves markedly on below and in_front and is competitive elsewhere.

Table 5: Per-relation accuracy (full-coverage SEA vs. baseline).

Relation	SEA	Baseline
below	0.84	0.56
in_front	0.75	0.54
on	0.83	0.80
right_of	0.83	0.80
behind	0.77	0.77
contains	0.60	0.61
left_of	0.73	0.76
above	0.78	0.85

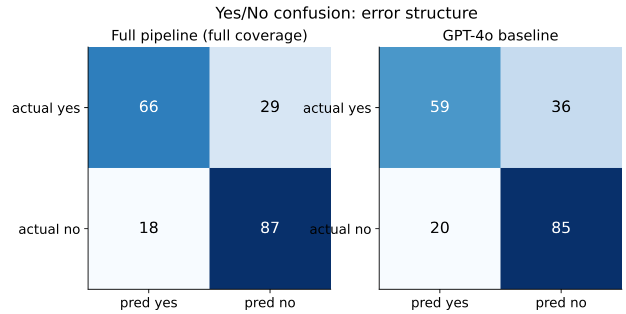


Figure 6: Yes/No confusion. SEA (left) recovers more true "yes" answers than the baseline (right) at similar false-positive cost.

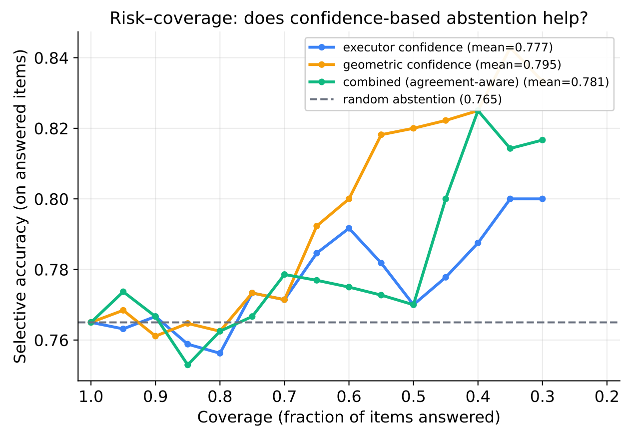


Figure 7: Risk–coverage. Sorting by geometric confidence and abstaining on the least-confident items raises selective accuracy well above random abstention; the VLM's own confidence is far less informative.

+10-point precision gain over the bare VLM, for applications where declining to answer beats a likely error. Figure 9 places our reported configurations and the bare-VLM baseline in the coverage/accuracy plane: all SEA points dominate the baseline, and lowering coverage along the geometric-confidence threshold trades coverage for verified, higher-precision answers.

8 Qualitative Results

Figure 10 shows representative verified outputs. The Critic localises both objects, the geometric rule fires with a clear margin, and the SEG records the full evidence. Left/right and above/below relations verify robustly in a single iteration; containment verifies via the coverage/area rule; depth scenes are handled conservatively, deferring to the VLM when the depth margin is small.

9 Discussion

Our results invite a reinterpretation of how deterministic perception modules should be combined with strong VLMs. The intuitive hope — that closed-form geometry can correct a VLM's spatial mistakes — holds only weakly: when high-confidence monocular geometry contradicts GPT-4o, the VLM is usually the more reliable party, particularly on depth. Overriding the VLM on such disagreements is therefore counterproductive, which is precisely why SEA's reconciliation policy defers depth disagreements rather than overruling them.

What geometry does provide is a trustworthy, model-independent estimate of when a spatial answer is safe. Detector confidence and rule-margin clearance jointly form a signal that ranks errors far better than the VLM's own (badly calibrated) confidence. SEA exploits this in two ways: agreement between the two independent estimators is a strong correctness cue (hence the high accuracy of the agreement bucket), and the geometric-confidence score gives a continuous knob for selective prediction. The net effect is a system that is both more accurate than the bare VLM at full coverage and fully auditable — every answer is accompanied by the boxes, depths, and rule evaluation that produced it.

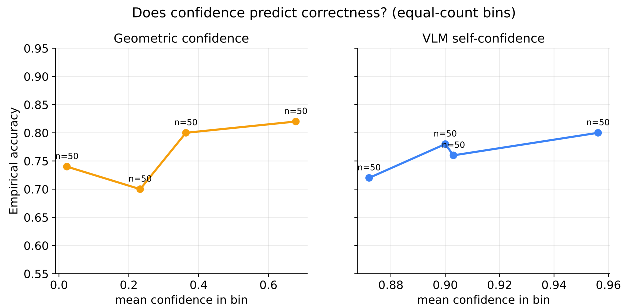


Figure 8: Empirical accuracy vs. confidence bin. Geometric confidence (left) predicts correctness; the VLM's self-confidence (right) is compressed near 1.0 and far less discriminative.

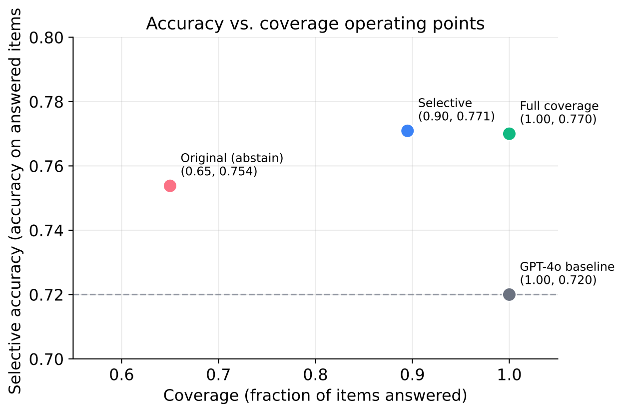
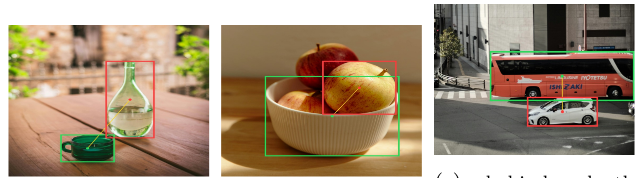


Figure 9: Operating points in the coverage/accuracy plane. Both SEA configurations sit above the GPT-4o baseline; the selective point answers 89.5% of items at higher selective accuracy.

This view generalises beyond spatial VQA. Any setting in which a strong but opaque generative model is paired with a weaker but transparent verifier can benefit from the same recipe: rather than forcing the verifier to overrule the generator, use the verifier's agreement and confidence to route and to abstain. The transparency is not free — it costs the extra detection and depth passes — but it converts an unaccountable black-box answer into one backed by inspectable evidence, which is precisely what safety-sensitive deployments require.

Future work. Three directions follow directly from our analysis. First, replacing relative monocular depth with a metric estimator would let the Critic safely correct the VLM on depth relations rather than only defer, potentially lifting the behind/in_front ceiling. Second, because geometric confidence is a calibrated error signal, an abstention threshold can be learned on held-out data to target a desired precision, replacing the hand-set τ. Third, passing VSR's reference-frame annotation to the Planner would let SEA distinguish



(a) left_of, verified (b) contains, verified (c) behind, depth scene

Figure 10: Annotated outputs. Green/red boxes are GroundingDINO detections for $\text{obj}_1/\text{obj}_2$; the applied rule and its value are stored in the SEG.

camera-relative from object-relative left/right queries, removing a known source of position-relation error. Finally, scaling the evaluation to the full VSR test set and to controlled benchmarks such as WhatsUp [5] would tighten the per-relation comparisons that are currently estimated at ∼25 samples each.

10 Limitations

Monocular depth. Relative single-image depth remains the weakest signal; SEA mitigates this by deferring depth disagreements to the VLM rather than overriding them, but metric depth would enable genuine geometric correction of depth relations. Scale. VSR-200 covers COCO scenes with ∼25 items per relation; per-relation comparisons are indicative, and broader evaluation across domains is future work. Reference frame. VSR mixes camera-relative and object-relative framings for left/right; SEA assumes the camera frame, accounting for some residual position-relation errors. Inference cost. SEA issues a Planner call, one or more Executor calls, and up to k local Critic passes per query, versus a single VLM call for the baseline; the full VSR-200 evaluation cost a few US dollars in API spend plus GPU time for detection and depth. SEA is therefore best suited to settings where verifiability and reliability outweigh per-query cost, rather than latency-critical real-time use.

11 Conclusion

We presented the Spatial Evidence Agent, a training-free three-agent pipeline that fuses VLM querying with deterministic geometric verification and a confidence-based reconciliation policy. On a 200-sample VSR split SEA reaches 77.0% accuracy at full coverage (+5.0 over a GPT-4o baseline) and 77.1% selective accuracy at 89.5% coverage with fully auditable evidence. Our analysis shows that the contribution of deterministic geometry is best understood as verification and uncertainty estimation: geometric confidence reliably indicates when a VLM's spatial answer is trustworthy, even where geometry alone is a weaker answerer. Future work includes metric-depth integration, reference-frame-aware predicates, and learned, calibrated abstention thresholds derived from the geometric-confidence signal.

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